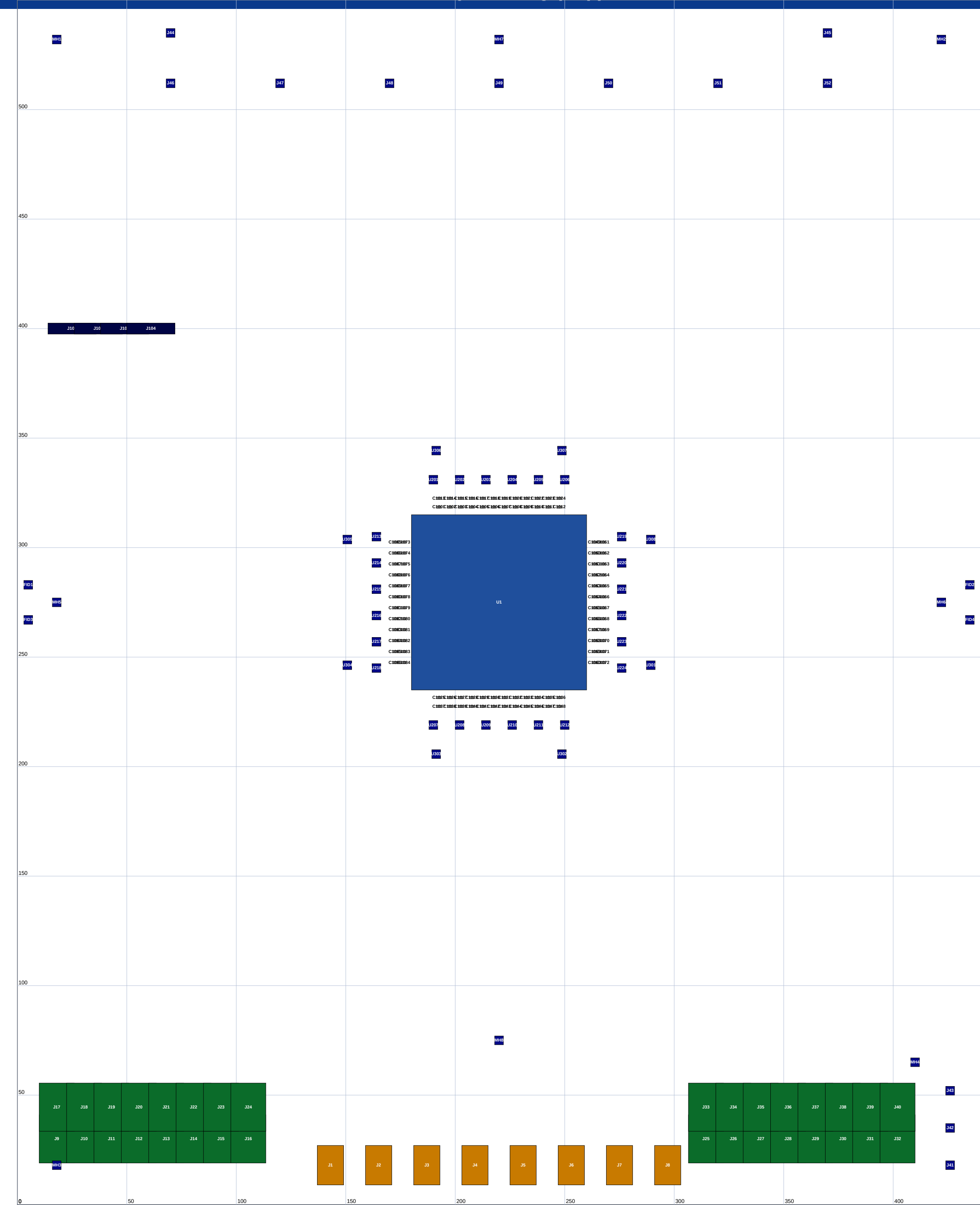


LR-S1A 1U CPO Switch Motherboard — Assembly Drawing (Top)



Reference-designator summary — Top side

Prefix	Function	Qty	Examples
C	Capacitors	96 pcs	C1001, C1002, C1003, C1004, C1005, C1006 ...
J	Connectors	56 pcs	J101, J102, J103, J104, J1, J2 ...
U	ASICs / silicon	33 pcs	U1, U201, U202, U203, U204, U205 ...
MH	Mounting holes	8 pcs	MH1, MH2, MH3, MH4, MH5, MH6 ...
FID	Fiducials	4 pcs	FID1, FID2, FID3, FID4

Assembly notes

2. SMT assembly per IPC-A-610 Class 3 - 100 % visual inspection.
3. Reflow 8-zone Pb-free SAC305 - peak 245 °C ± 50 °C - TAL 60 ± 10 s.
4. LR-CSW-51T2 484-ball BGA - placement ± 50 µm - 100 % AXI - 100 % first-article X-ray.
4. Underfill LR-CSW component (Loctite 3593) - corner-bond LR-NCE daughter ICs.
5. ESD per ANSI/ESD-S20.20 - HBM 2 kV / CDM 500 V minimum.
6. Hand-assembly: ELSPF gases, OSPF-XD gases, MPO inlets, cold-plate, PSU bays.
7. Cold-plate retention 4xM2.5 to 0.5 N·m cross pattern - TIM Honeywell PTM7950.
8. Fan headers keyed - install fans after PSU bring-up ± 36 mm H₂O at 12 000 rpm.
9. PSU bay 25-cycle hot-swap - CRPS pin-mate alignment ± 0.3 mm.
10. Cleaning aqueous DI + saponifier - ≤ 1.56 µg NaCl/cm² (IPC-TM-650 2.3.25).
11. No conformal coating (liquid-cooled, sealed chassis).
12. Hi-pot 500 VDC 60 s between V_PSU_12V_A/B and GND - ≤ 1 mA leakage.
13. ICT coverage ≥ 95 % nets - functional CPU + BMC + ASIC PLL + HBM init test.
14. Data code on D.Silk - IPC-7351B 2D-data-matrix - YYYW format.
15. QC: 24 h 65 °C burn-in - full PDN load - fan speed - 32-port link-up.

LightRail AI — Confidential / Investor data-room	
Programme	LR-S1A 1U CPO Switch Motherboard
Drawing No.	LR-S1A-ASM-001
Title	Assembly Drawing — Top side
Material	18-layer HDI · Megtron-7 / FR-4 / BC24
Designer	LightRail AI — Design Automation
Date / Rev	2026-04-19 · Rev 1.0